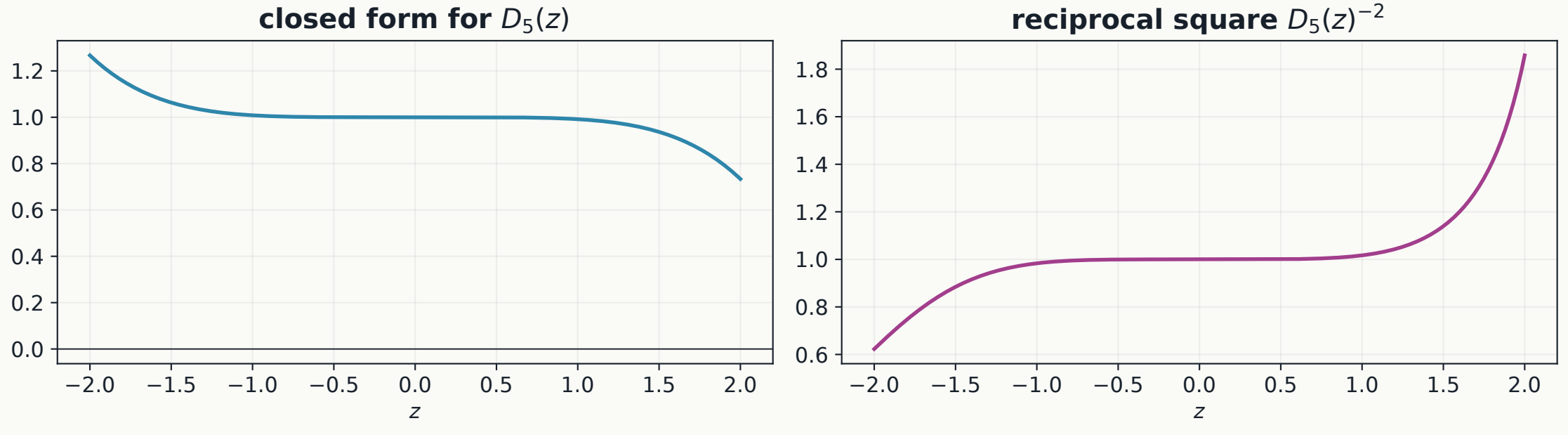A root-of-unity packet sum produces an explicit analytic family